Paper 2

Additional Mathematics 4049/02

October/November 2024
2 hours 15 minutes

*Answer **all** the questions.*

1. The manager of a coffee shop purchases a coffee machine for \$1800. The value of the machine, \$$V$, after a period of t months can be modelled by the formula $V = 1800e^{-kt}$ where k is a constant. After 12 months, the value of the machine has dropped to \$1000. The manager purchases a new machine when the value of the first machine is 25% of its purchase price. Find the value of t when a new machine is purchased. [4]

2. Find the range of values of the constant k for which the line $y = kx$ intersects the curve $y = x^2 + 3kx + 2 - k$ at two distinct points. [5]

3. Solve the equation $\log_2 x - \log_4 (x + 3) = 3 \log_8 2$. [5]

4. **(a)** **(i)** Find the gradient of the curve $y = \dfrac{4x}{5 - 3x}$ at the point where $x = 1$. [3]

(ii) Explain why the curve has no stationary points. [1]

(b) The curve $y = f(x)$ is such that $f'(x) = \dfrac{4}{\cos^2 x}$. The curve passes through the point $\left(\dfrac{\pi}{4}, 1\right)$. Find an expression for $f(x)$. [3]

5. The equation of a curve is $y = e^x + e^{-x}$.

(a) Verify that the curve has a stationary point at $x = 0$. [2]

(b) Solve the equation $e^x + e^{-x} = 2.9$, giving values of x in logarithmic form. [4]

6. (a) The expressions $x^3 + x^2 + ax + 2b$ and $x^3 + 4x^2 - ax + b$, where a and b are constants, both have a factor of $x + 3$. Find the values of a and b. [4]

(b) Solve the equation $x^3 + 2x^2 - 11x + 6 = 0$, expressing the roots in an exact form. [5]

7. Points $P(3, 1)$ and $Q(-5, 7)$ lie on a circle and PQ is a diameter of the circle.

(a) Find the coordinates of the centre of the circle and the radius. [3]

(b) Find the equation of the circle in the form $x^2 + y^2 + 2gx + 2fy + c = 0$. [2]

The point R lies on the line $3y = 4x - 34$ and is the point on the line that is closest to the circle.

(c) Without finding the coordinates of R, explain why R lies on QP extended. [1]

(d) Hence find the coordinates of R. [3]

8. At time $t = 0$, where t is measured in seconds, a racing car comes out of a bend at a point P with a speed of 25 m/s. The car then travels in a straight line and when $t = 4$ it passes a point Q with a speed of 85 m/s. For the motion from P to Q, its speed, v m/s, can be modelled by the formula $v = At^{2.5} + 11t + C$, where A and C are constants.

(a) Explain why $C = 25$ and find the value of A. [3]

(b) Find the acceleration of the car as it passes Q. [2]

(c) Find the distance PQ to the nearest metre. [4]

9. **(a)** Prove the identity $\tan x + \cot x = 2 \operatorname{cosec} 2x$. [4]

(b) Solve the equation $\tan x + \cot x = 3$ for $0 \leqslant x \leqslant 2\pi$. [6]

(c) Use the result in **part (a)** to find the set of values of the constant k for which there are no solutions to the equation $\tan x + \cot x = k$. [2]

10. (a) When an object falls, it experiences air resistance. For one particular object, the resistance, R, depends upon the velocity, v, of the object and can be modelled by the formula $R = Av^k$, where A and k are constants. The table below shows corresponding values of v and R. It is believed that an error was made in recording one of the values of R.

v	5	10	15	20	25
R	21	55	110	146	200

(i) Explain how a straight line graph can be drawn to represent the formula and draw it for the given data. [4]

(ii) From your straight line graph, identify the incorrect reading and suggest a corrected value for R. [2]

(iii) Estimate the value of k. [2]

(b) Variables x and y are related by the equation $y = ax + \dfrac{b}{x}$, where a and b are constants. Explain clearly how a straight line graph can be drawn to represent this equation. You should state which variables should be plotted on each axis and explain how the values of a and b can be calculated. [4]

11. The equation of a curve is $y = e^{-2x} \sin x$.

(a) Explain why the curve lies completely above the x-axis for $0 < x < \pi$. [2]

(b) Find $\frac{dy}{dx}$ and show that the x-coordinate of the stationary point for $0 < x < \pi$ is $\tan^{-1}\frac{1}{2}$. [5]

(c) Find $\frac{d^2y}{dx^2}$ and show that $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 5y = 0$. [5]